EXPERIMENT-3

NAME: D.SARVESHVARN ROLL.NO: 240701480

AIM: To design and implement an Expense Tracker application using Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) in Python, and to compare their usability.

1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
expenses = []

def add_expenses():

    category = input("Enter category (FOOD/TRAVEL/STUDY/OTHER): ")

    amount = int(input("Enter amount: "))

    expenses.append((category, amount))

    print("Expense added successfully")


def view_expenses():

    if not expenses:

        print("\nNo expenses added yet.")

        return

    print("\nExpenses:")

    for cat, amt in expenses:

        print(f"{cat} - ₹{amt}")


while True:

    print("\n1. Add Expenses")
```

```
print("2. View Expenses")
print("3. Exit")

choice = input("Enter choice: ").strip()

if choice == "1":
    add_expenses()
elif choice == "2":
    view_expenses()
elif choice == "3":
    print("\nThank You!!!!")
    break
else:
    print("Invalid Choice")
```

OUTPUT:

```
PS C:\Users\sarxe> & "C:/Program Files/Python312/python.exe" c:/Users/sarxe/OneDrive/Desktop/expt-3.py

1. Add Expenses
2. View Expenses
3. Exit
Enter choice: 1
Enter category (FOOD/TRAVEL/STUDY/OTHER): food
Enter amount: 1000
Expense added successfully

1. Add Expenses
2. View Expenses
3. Exit
Enter choice: 2

Expenses:
food  - ₹1000

1. Add Expenses
2. View Expenses
3. Exit
Enter choice: 3

Thank You!!!!
```

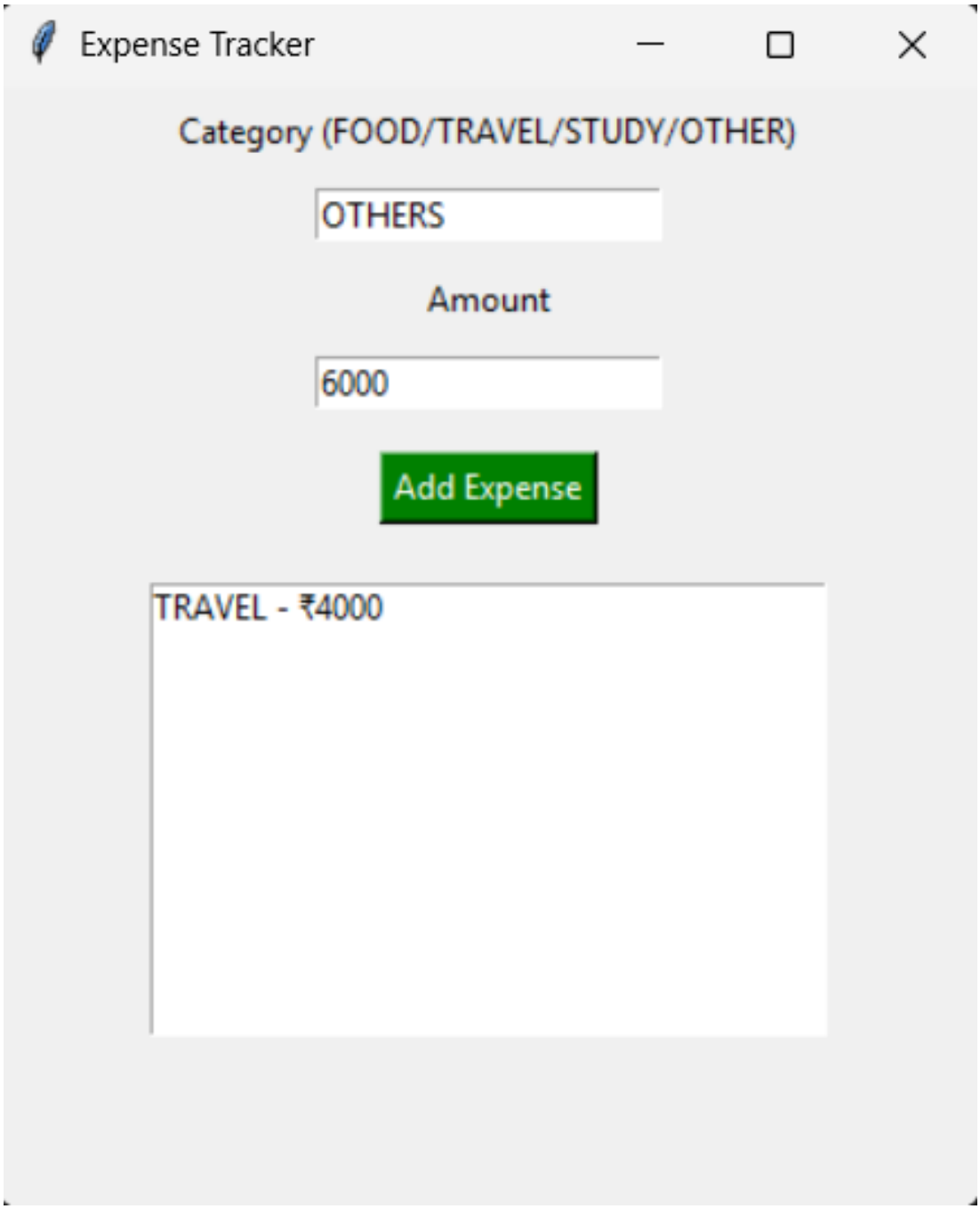
2. GRAPHICAL USER INTERFACE:

PYTHON CODE:

```
import tkinter as tk
from tkinter import messagebox
expenses = []

def add_expense():
    category = category_entry.get()
    amount = amount_entry.get()
    if category == "" or amount == "":
        messagebox.showwarning("Warning", "Please enter all fields")
        return
    expenses.append((category, amount))
    expense_list.insert(tk.END, f"{category} - ₹{amount}")
    category_entry.delete(0, tk.END)
    amount_entry.delete(0, tk.END)

root = tk.Tk()
root.title("Expense Tracker")
root.geometry("350x400")
tk.Label(root, text="Category (FOOD/TRAVEL/STUDY/OTHER)").pack(pady=5)
category_entry = tk.Entry(root)
category_entry.pack(pady=5)
tk.Label(root, text="Amount").pack(pady=5)
amount_entry = tk.Entry(root)
amount_entry.pack(pady=5)
tk.Button(root, text="Add Expense", command=add_expense, bg="green",
fg="white").pack(pady=10)
```

```
expense_list = tk.Listbox(root, width=40, height=10)

expense_list.pack(pady=10)

root.mainloop()
```

OUTPUT:



3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr
expenses = []

def process_command(command):
    words = command.lower().split()
    if len(words) == 4 and words[0] == "add" and words[1] == "expense":
        category = words[2]
        amount = words[3]
        expenses.append((category, amount))
        print(f"Expense added: {category} - ₹{amount}")
    else:
        print("Invalid command format")

recognizer = sr.Recognizer()
mic = sr.Microphone()
print("🎙 Say: add expense food 120")
with mic as source:
    recognizer.adjust_for_ambient_noise(source)
    audio = recognizer.listen(source)

try:
    command = recognizer.recognize_google(audio)
    print("You said:", command)
    process_command(command)
except:
```

```
    print("Sorry, could not understand")
```

OUTPUT:

```
PS C:\Users\sarxe> & "C:/Program Files/Python312/python.exe" "c:/Users/sarxe/OneDrive/Desktop/import tkinter as tk.py"
Say: add expense food 120
You said: add expense travel 1500
Expense added: travel - ₹1500
PS C:\Users\sarxe>
```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def user_satisfaction():
    print("Rate your satisfaction with the following interfaces (1-5):")
    try:
        cli = int(input("CLI (1-5): "))
        gui = int(input("GUI (1-5): "))
        vui = int(input("VUI (1-5): "))
        if not (1 <= cli <= 5 and 1 <= gui <= 5 and 1 <= vui <= 5):
            print("Please enter ratings between 1 and 5 only.")
            return

        print("\nUser Satisfaction Ratings:")
        print(f"CLI: {cli}")
        print(f"GUI: {gui}")
        print(f"VUI: {vui}")
```

```
        average = (cli + gui + vui) / 3
        print(f"\nAverage Satisfaction Score: {average:.2f}")
    except ValueError:
        print("Invalid input! Please enter numbers only.")


if __name__ == "__main__":
    user_satisfaction()
```

OUTPUT:

```
PS C:\Users\sarxe> & "C:/Program Files/Python312/python.exe" "c:/Users/sarxe/OneDrive/Desktop/import tkinter as tk.py"
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 3

User Satisfaction Ratings:
CLI: 4
GUI: 5
VUI: 3

Average Satisfaction Score: 4.00
PS C:\Users\sarxe>
```